QUOTATION

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Robotic Spectrometer

DELIVERY SET AND PRICE

Item	Item price, EUR	Q-ty	Unit price, EUR
Gamma detection unit	94 500.00	1	94 500.00
NaI 3x3 scintillation block with well type			
Cable set with connectors			
Electronics			
Detachable spectrometry electronics unit with LAN (MCA Base)			
Software packages			
Analytical software GammaPRO			
Sample management software			
Special purpose software			
Automatic sample counter based on Robot arm			
Robot arm with controller			
Table for sample placement			
Safety Housing protection			
Barcode printer and barcode scanner			
Low background lead shielding			
Side sample rack lead shielding			
Modern and up-to-date working station			
		Total price, EUR	**94 500.00**
Options			
Installation commissioning and training (visit of 1-2 specialist for 10-11 working days)**	+17 000.00	1	+ 17 000.00
Calibration Standards Cs-137 and Co-60	+ 6 800.00	1	+ 6 800.00
Detector characterization	+ 3 000.00	1	+ 3 000.00
Working Station	+ 2 500.00	1	+ 2 500.00

TERMS AND CONDITIONS

Delivery conditions	FCA, Riga
Payment terms:	Wire transfer, prepayment
Delivery terms:	9-12 months ARO *
Warranty on labor and parts:	1 year
Quotation validity:	4 months
Country of Origin	Latvia

** - to be specified in case of order.*

***- Expected assistance from the End-User (unpacking and operational activities)*

ROBOTIC SPECTROMETER SYSTEM BENEFITS

Automated Throughput
Robotic processing of 250+ samples enables unattended gamma measurements and efficient laboratory workflows.

Safer Sample Handling
Automatic transfer between sample rack and shielded detector reduces manual handling of radioactive materials.

Integrated Sample-to-Result Workflow
Robotic handling combines sample identification, automatic weighing and gamma measurement into one controlled sequence.

Measurement Traceability
Sample identification, weight, spectrum and activity results are linked and stored for reliable documentation and reporting.

24/7 Unattended Operation
Designed for continuous day-and-night operation, the robotic system can process scheduled sample batches without an operator being continuously present, maximizing instrument utilization and laboratory productivity.

APPLICATION

Scintillation Gamma Spectrometer-Radiometer, is designed for detection of, respectively, gamma radiation, measuring specific and volumetric activity of natural and artificial radionuclides in soil samples, rocks, vegetation, water, food, wood, building materials, chemical industry materials, radiopharmaceuticals, alloys, scrap metal and other technological products.

TECHNICAL SPECIFICATION OF BDEG-76-76

Description	Value
Detector type: - Scintillation well-type gamma ray detector	NaI(Tl) Ø76x76 mm
Energy resolution at 661keV for detector: -	< 7.5%
Detectable energy range, keV - gamma radiation	40-3000
Maximum throughput, cps, at least	$5 \cdot 10^4$
No. of channel for MCA, at least	2048- MCA Channels
Instability of spectrometer indications (energy conversion characteristic) for 8 hours of a continuous operation, %	±1
Operation conditions: • ambient air temperature, °C • relative air humidity, % • atmospheric pressure in the range of, kPa • intensity of magnetic fields of the permanent and variable grid frequencies, A/m	• from +0 to +40 • up to 90 at no condensation • from 84 to 107 • up to 40

Analytical software GammaPRO

GammaPRO provides all the tools necessary for separate peak analysis, namely:

- Energy calibration;
- Direct peak search in spectrum and peak search considering the radionuclides and their peak energies specified in the library;
- Peak and multiplet fit by Gaussian and other functions;
- Automatic identification of peaks;
- Considering count sample density as a calculation parameter; considering background as a calculation parameter;
- Calculation of efficiency curves (efficiency calibration), FWHM calibration, peak shape calibration;

GammaPRO software enables solution of specific spectrometric problems such as:

- Calculation of activity of naturally occurring radionuclides and effective specific activity in situ and in samples calculation of conformity factors for food and materials;
- Calculation of radionuclide activity applying methods of physical and radiochemical concentrating;

GammaPRO software features multiple window interface and enables a number of opportunities for spectrum analysis such as:

- Visualization of spectra and spectrum acquisition progress;
- Peak search and identification of radionuclides;
- Calculation of efficiency curves and sensitivities;
- Mathematical operations (sum, subtraction, normalization, etc.);
- Processing of set of spectra;
- Calculation of integral nonlinearity;
- Conversion into other formats and translation into other applications such as MS Word, Excel, MS Access, MatLab;

LEAD SHIELD Low background lead shield chamber

The spectrometer is mounted in a compact shield chamber featuring a thickness of the lead layer of ca 5 cm on the stable platform. The shield chambers are used to protect measuring sample from interactions with other samples, laboratory or natural background that is especially essential in case of acquiring spectrum of low activity samples.

Cylindrical top-opening lead shield:

- 50 mm thickness of the lead wall;
- Additional shielding of 75 mm lead from the side sample rack;

AUTOMATIC SAMPLE CHANGER

The seven-axis robot handles a payload of up to 3kg, and with, practically, unlimited reach, the robot is able to carry out a series of operations using flexible rather than hard automated solutions. In addition to a horizontal reach, the robot has the ability to reach below its base. Furthermore, the robot has a very compact turning radius, which is enabled by the robot's symmetric architecture, without offset on axis 2. This ensures the robot can be mounted close to other equipment.
The Robotic spectrometer is placed on the aluminum frame made of profile, equipped with protective cabinet with plastic windows and protective sensors indicating if the door is open.
The size of the aluminum profile frame is approximately 1500 mm x 1200 mm x 1500 mm.

**The exact dimensions of the aluminum profile frame depend on the quantity and size of the samples and will be specified when placing an order.*

ROBOT ARM BASIC CHARACTERISTICS:



- Seven-axis manipulator
- Machine vision
- Payload: up to 1 kg
- Reach: per request
- Fastest 7-axis robot
- Accuracy: +-0.01 mm
- IP30 protected
- All motors and cablings enclosed
- Compact controller
- Sample holder tool for vessels with diameter in range 13 – 28 mm.
- Barcode Reader and Writer

To assure the correct processing of all data during the measurement and analysis process, the samples are marked by using a barcode printer that is connected to the workstation. Here all necessary information about the sample is stored in a database. Using the bar code reader, the information stored in the database is retrieved for each sample before the measurement process is started. This fully automated process delivers all necessary information for the measurement and analysis process.

Compact controller

The Compact controller delivers space saving benefits and easy commissioning through one phase power input, external connectors for all signals. It supports the state-of-the-art field busses for I/O. Sensor interface functionality, remote disk access and socket messaging are examples of the many powerful networking features. Remote monitoring of the robot is available through standard communication network (Ethernet). Advanced diagnostic methods allow fast investigation on failure as well as monitoring of the robot condition throughout the life cycle.

Barcode printer and Barcode reader

Printer is located outside of frame and intended for barcode printing and manual sticking of label on the sample before fixture is loaded inside. Reader is located inside the frame for automatic reading the sample information before loading into lead chamber.

Samples table

Samples table is designed to accommodate 250+ samples. Sample size and amount directly influence on table size. Special holding tool of Robot arm can be designed to hold:

- Test tube;

** To be specified by the end user: the geometry and size of the sample.*

Options

Advanced software package MCC-MT

Software package MCC-MT (Monte Carlo Calculation Multi Thread) is intended for 3D-modelling of physical experiments and calculation of radiation detectors response functions using Monte Carlo simulation method. Software MCC-MT based on multi-threading technology providing significant increasing the rate of simulation and getting fast result as spectrum.

Application

- Monte Carlo simulation spectra of gamma, beta and alpha radiation
- calibration of instruments used for ionizing radiation detection and measurements
- calculation of detection limits and minimum detectable activity of radionuclides
- determination of a characteristics of a registration system for inaccessible radioactive sources
- reduction of experimental investigations with using the hazardous ionizing radiation for human health
- obtaining clear picture of the internal processes of radiation transfer in order to optimize the design of the measuring devices and their protection
- comparative demonstration of the different systems of protection against ionizing radiation and its detection systems
- training of personnel in working with ionizing radiation detection systems without using of an expensive equipment and radioactive sources
- training of specialists in the field of measurement and protection from ionizing radiation
- acceleration, simplification and reduction in the cost of design and optimization of ionizing radiation detection systems
- characterization detectors and detection systems

Features

- high accuracy of calculations
- simplicity of using for a wide range of tasks
- detailed 3D-scene based on Open GL graphics technology providing maximum representation and visibility of modeling
- availability of replenished database of sources and materials
- possibility of creating the maximally complex measuring systems
- forming multidetector systems and schemes of coincidence
- display of the results in the form of an ideal and real spectrum
- tracing and drawing trajectories of particles during calculation process
- availability of the ready and test projects in the software package
- accounting cascade summation ('Full cascade' source type)
- using of the specified number of computer processors in order to implement multithreading and speed up the calculation
- automatic creation of efficiency curves

Workstation – PC based*

• 24 inch LCD Monitors • CPU : core I-7 or better • RAM : 8GB DDR3 or better • HDD : 1 TB or better • Uninterruptable Power Supply (UPS) for 30 minute backup(except for robotic system)	• ODD : SUPER-MULTI S-ATA or better • OS : WINDOWS 11 • Laser Printer • RAID system

**Please note that these values are for reference only. Precise specification may change to more advanced system.*